Machine Learning

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Linear Regression

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Introduction

- Linear Regression is a fundamental supervised learning algorithm used for predicting continuous outcomes.
- It models the relationship between a dependent variable and one or more independent variables using a linear equation.
- The goal is to find the best-fitting line that minimizes prediction error.
- It forms the basis of many advanced machine learning techniques.

Real-world Applications:

- Predicting house prices based on area and location
- Estimating salary using experience
- Sales forecasting using advertising expenditure
- Weather and temperature prediction

Types of Linear Regression

1. **Simple Linear Regression**
   - Uses one independent variable
   - Models a straight-line relationship
   - Example: Salary vs Experience

2. **Multiple Linear Regression**
   - Uses multiple independent variables
   - Captures complex relationships
   - Example: House price using area, location, rooms

Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + \epsilon$$

- y = Dependent variable
- x = Independent variable
- β_0 = Intercept
- β_1 = Slope
- ϵ = Error term

Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n + \epsilon$$

- Multiple predictors influence output
- More realistic for real-world problems
- Each coefficient represents variable contribution

Goal of Simple Linear Regression

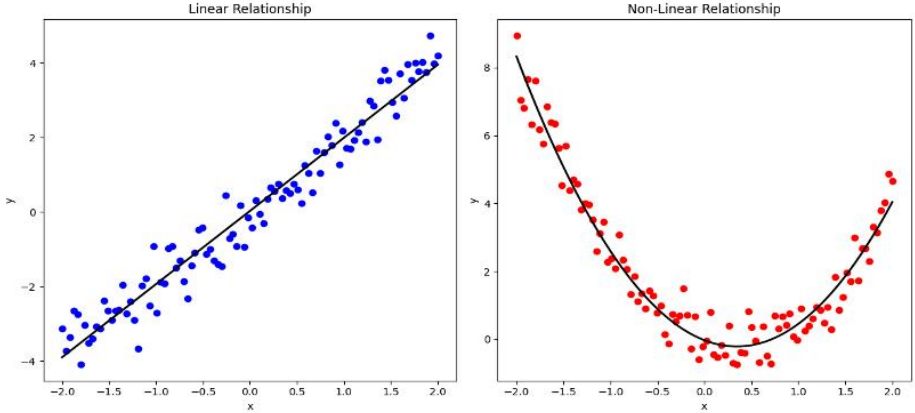
- Find the best-fitting line through data
- Minimize prediction error
- Estimate optimal parameters β_0 and β_1
- Ensure good generalization

Assumptions of Linear Regression

- **Linearity:** Linear relationship between variables
- **Independence:** Observations are independent
- **Homoscedasticity:** Constant variance of errors
- **Normality:** Residuals are normally distributed
- **No Multicollinearity:** Predictors are not highly correlated

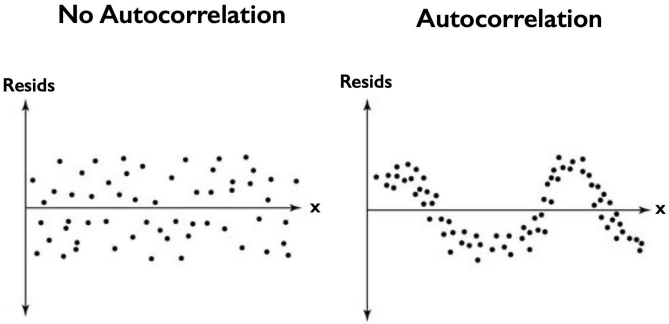
1. Linearity

- Straight-line relationship between X and Y
- Non-linearity reduces model accuracy
- Checked using scatter plots



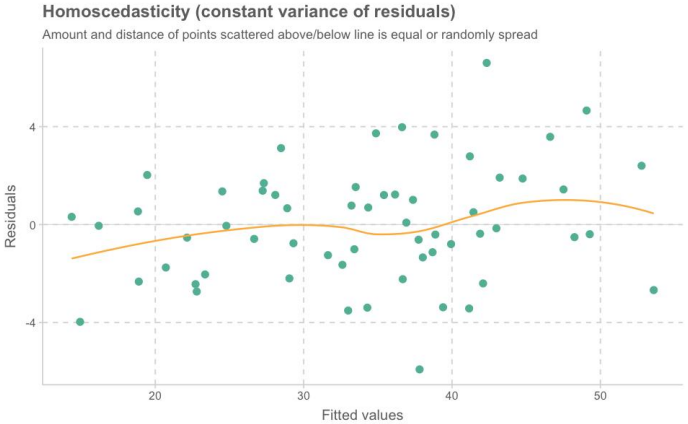
2. Independence of Errors

- Errors should not be correlated
- Important in time-series data
- Violation leads to biased results



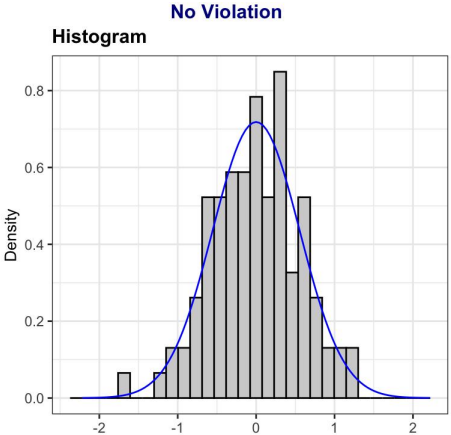
3. Homoscedasticity

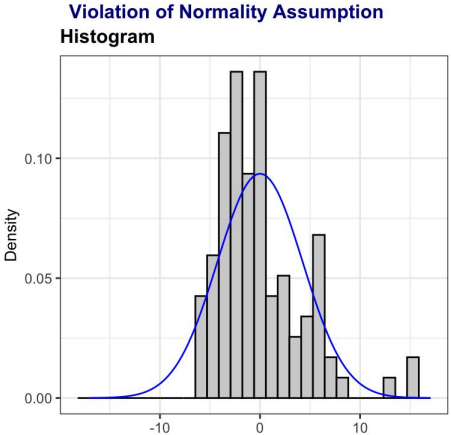
- Constant variance of residuals
- Funnel shape indicates violation
- Affects reliability



4. Normality of Errors

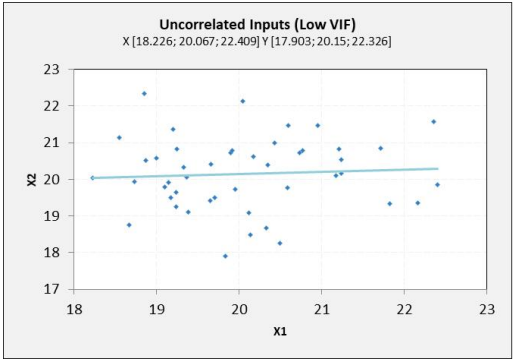
- Residuals should follow normal distribution
- Important for statistical inference
- Checked via histogram or Q-Q plot

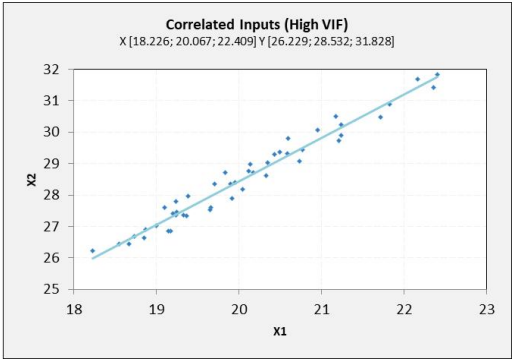




5. No Multicollinearity

- Predictors should not be highly correlated
- Causes unstable coefficients
- Detected using VIF





Example

$$y = 5 + 2x$$

- $\beta_0 = 5$ (Intercept)
- $\beta_1 = 2$ (Slope)
- When $x = 0$, $y = 5$
- Increase of 1 in x increases y by 2

Ordinary Least Squares (OLS)

Objective: Estimate parameters by minimizing the sum of squared errors.

$$\text{Minimize} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- OLS finds the best-fitting line by minimizing squared residuals
- Residual = Difference between actual and predicted value
- Provides closed-form solution for parameters

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

OLS Derivation

Step 1: Define the Cost Function

$$J(\beta_0, \beta_1) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

- Measures total squared error between actual and predicted values
- Goal: Find β_0 and β_1 that minimize J
- Factor $\frac{1}{2n}$ simplifies derivative calculations

OLS Derivation

Cost Function:

$$J(\beta_0, \beta_1) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

Expand the squared term:

$$= \frac{1}{2n} \sum_{i=1}^{n} \left(y_i^2 - 2y_i(\beta_0 + \beta_1 x_i) + (\beta_0 + \beta_1 x_i)^2 \right)$$

- Converts problem into differentiable form
- Enables analytical minimization

Derivative w.r.t β_0

$$\frac{\partial J}{\partial \beta_0} = \frac{1}{2n}\sum_{i=1}^{n} 2(y_i - (\beta_0 + \beta_1 x_i))(-1)$$

$$= -\frac{1}{n}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)$$

Set derivative to zero:

$$\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i) = 0$$

Derivative w.r.t β_1

$$\frac{\partial J}{\partial \beta_1} = \frac{1}{2n}\sum_{i=1}^{n} 2(y_i - (\beta_0 + \beta_1 x_i))(-x_i)$$

$$= -\frac{1}{n}\sum_{i=1}^{n} x_i(y_i - \beta_0 - \beta_1 x_i)$$

Set derivative to zero:

$$\sum_{i=1}^{n} x_i(y_i - \beta_0 - \beta_1 x_i) = 0$$

Solving the Equations

From $\frac{\partial J}{\partial \beta_0} = 0$:

$$\sum y_i = n\beta_0 + \beta_1 \sum x_i$$

From $\frac{\partial J}{\partial \beta_1} = 0$:

$$\sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2$$

These are called Normal Equations

Final OLS Solution

Solving the normal equations:

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

- Gives optimal parameters
- Minimizes squared error globally
- Unique solution if variance of X is non-zero

Closed-Form Solution (Matrix Form)

$$\beta = (X^T X)^{-1} X^T y$$

- X = Feature matrix
- y = Target vector
- Provides closed-form solution for multiple regression
- Requires $X^T X$ to be invertible

Closed-Form Solution (Derivation)

Linear Regression Model:

$$\hat{y} = X\beta + \epsilon$$

- X = Feature matrix $(n \times p)$
- β = Parameter vector $(p \times 1)$
- $\hat{y}$ = Output vector $(n \times 1)$

Objective:

$$\text{Minimize } \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

can be writeen in matrix Form as:

$$\min_{\beta} (y - X\beta)^T (y - X\beta)$$

Expand the Cost Function

$$J(\beta) = (y - X\beta)^T (y - X\beta)$$

Expand:

$$= y^T y - y^T X\beta - (X\beta)^T y + (X\beta)^T (X\beta)$$

$$= y^T y - 2\beta^T X^T y + \beta^T X^T X\beta$$

- Uses property: $(X\beta)^T = \beta^T X^T$
- Converts into differentiable quadratic form

Derivative w.r.t β

$$\frac{\partial J}{\partial \beta} = -2X^T y + 2X^T X\beta$$

Set derivative to zero:

$$-2X^T y + 2X^T X\beta = 0$$

$$X^T X\beta = X^T y$$

So,

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T y$$

Applications

- House price prediction
- Sales forecasting
- Risk analysis
- Economic modeling

Thank You